

Best Augmenting Combos for Partial Response or Treatment Resistance

- Generally, best to switch to another agent
- Eszopiclone, zolpidem
- Trazodone
- Agents with antihistamine actions (e.g., diphenhydramine, TCAs)

Tests

- None for healthy individuals
- For patients presenting with unexplained amenorrhea, galactorrhea, decreased libido, or problems with fertility, could consider measuring prolactin and testosterone levels

Side Effects

How Drug Causes Side Effects

- Actions at melatonin receptors that carry over to the next day could theoretically cause daytime sedation, fatigue, and sluggishness, but this is not common

Notable Side Effects

- * Sedation
- * Dizziness

* Fatigue

* Headache



Life-Threatening or Dangerous Side Effects

- Respiratory depression, especially when taken with other CNS depressants in overdose
- Rare angioedema

Weight Gain



- Reported but not expected

Sedation



- Many experience and/or can be significant in amount
- May experience sedation or sleepiness immediately after dosing, but not commonly after awakening from a night's sleep

What to Do About Side Effects

- Wait
- To avoid problems with memory, only take ramelteon if planning to have a full night's sleep
- Lower the dose

- Switch to a non-benzodiazepine sedative hypnotic

Best Augmenting Agents for Side Effects

- Many side effects cannot be improved with an augmenting agent

Dosing and Use

Usual Dosage Range

- 8 mg at bedtime

Dosage Forms

- Tablet 8 mg

How to Dose

- No titration, take dose at bedtime



Dosing Tips

- Unusual lack of apparent dose response curve
- Doses between 4 mg and 64 mg may have similar effects on sleep and similar side effects
- Doses up to 160 mg were studied without apparent abuse liability
- Suggests therapeutic effects may be mediated by an “on-off” type of therapeutic effect on a sleep switch that works at any dose over a certain threshold

- Since ramelteon has very low oral bioavailability and thus highly variable absorption, a substantial dose range may be required to generate sufficient absorption in various patients
- Thus, “one size does not fit all” despite approval at only one dose without titration (i.e., 8 mg)
- Suggest increasing dose before concluding lack of efficacy
- Do not administer with or immediately after a high-fat meal as this may delay its onset of action or diminish its efficacy

Overdose

- No reports of ramelteon overdose
- No safety or tolerability concerns in studies up to 160 mg

Long-Term Use

- No reports of dependence, tolerance, or abuse liability
- Not restricted to short-term use but few long-term studies

Habit Forming

- No

How to Stop

- No evidence of rebound insomnia the first night after stopping
- No need to taper dose

Pharmacokinetics

- Metabolized predominantly by CYP450 1A2
- CYP450 3A4 and 2C are also involved in metabolism of ramelteon
- Mean elimination half-life of parent drug 1–2.6 hours
- Mean elimination half-life of major metabolite, M-II, is 2–5 hours



Drug Interactions

- Inhibitors of CYP450 1A2, such as fluvoxamine, could increase plasma levels of ramelteon
- Inducers of CYP450, such as rifampin, could decrease plasma levels of ramelteon
- Inhibitors of CYP450 3A4, such as ketoconazole, could increase plasma levels of ramelteon
- Inhibitors of CYP450 2C9, such as fluconazole, could increase plasma levels of ramelteon
- Exercise caution if combining with alcohol
- No interaction with fluoxetine (CYP450 2D6 inhibitor)



Other Warnings/Precautions

- Insomnia may be a symptom of a primary disorder, rather than a primary disorder itself
- Use only with extreme caution in patients with impaired respiratory function or obstructive sleep apnea

- Ramelteon should only be administered at bedtime
- May decrease testosterone levels or increase prolactin levels, but the clinical significance of this is unknown
- Patients who develop angioedema after treatment with ramelteon should not be rechallenged

Do Not Use

- With fluvoxamine
- In patients with severe hepatic impairment
- If there is a proven allergy to ramelteon

Special Populations

Renal Impairment

- Dose adjustment not generally necessary

Hepatic Impairment

- Use with caution in patients with moderate hepatic impairment
- Not recommended for use in patients with severe impairment

Cardiac Impairment

- Dosage adjustment may not be necessary

Elderly

- No adjustment necessary
- Greater absorption and higher plasma drug concentrations but no increase in side effects

Children and Adolescents

- Safety and efficacy have not been established

Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women

Breast Feeding

- Unknown if ramelteon is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- * Recommended either to discontinue drug or bottle feed

The Art of Psychopharmacology

Potential Advantages

- Those who require long-term treatment
- Those who need a hypnotic but have a past history of substance abuse
- Possibly for circadian rhythm disturbances

Potential Disadvantages

- More expensive than some other hypnotics
- For patients who require sedation

Primary Target Symptoms

- Time to sleep onset

Pearls

- First in a new class of agents, chronohypnotics, that act upon circadian rhythms by stimulating melatonin receptors in the brain's "pacemaker," namely the suprachiasmatic nucleus
- Theoretically, stimulation of melatonin 1 receptors mediates the suppressive effects of melatonin on the suprachiasmatic nucleus
- Theoretically, stimulation of melatonin 2 receptors mediates the phase shifting effect of melatonin
- Ramelteon may act by promoting the proper maintenance of circadian rhythms underlying a normal sleep-wake cycle

- Thus, ramelteon may also prove effective for treatment of circadian rhythm disturbances such as shift work sleep disorder and jet lag
- Lack of actions on GABA systems, which may be related to lack of apparent abuse liability
- Only approved hypnotic agent that is not scheduled and is considered to have no abuse liability
- No evidence that ramelteon worsens apnea/hypopnea index in COPD or in obstructive sleep apnea, but not recommended in severe cases
- * May be preferred over benzodiazepines because of its rapid onset of action, short duration of effect, and safety profile
- Rebound insomnia does not appear to be common
- May have fewer carryover side effects than some other sedative hypnotics

Suggested Reading

Erman M , Seiden D , Zammit G , Sainati S , Zhang J . An efficacy, safety, and dose-response study of ramelteon in patients with chronic primary insomnia . *Sleep Med* 2006 ;**7** (1):17 –24 .

Kato K , Hirai K , Nishiyama K , et al. Neurochemical properties of ramelteon (TAK-375), a selective MT1/MT2 receptor agonist . *Neuropharmacology* 2005 ;**48** (2):301–10 .

.....
McGechan A , Wellington K . Ramelteon . *CNS Drugs* 2005 ;**19** (12
):1057–65 .
.....

Pandi-Perumal SR , Zisapel N , Srinivasan V , Cardinali DP . Melatonin
and sleep in aging population . *Exp Gerontol* 2005 ;**40** (12):911–25 .

Reboxetine

Therapeutics

Brands

- Norebox
- Edronax
- *see index for additional brand names*

Generic?

No

Class

- Neuroscience-based Nomenclature: norepinephrine reuptake inhibitor (N-RI)
- Selective norepinephrine reuptake inhibitor (NRI); antidepressant

Commonly Prescribed for

(bold for FDA approved)

- Major depressive disorder
- Dysthymia
- Panic disorder
- Attention deficit hyperactivity disorder (ADHD)

How the Drug Works

- Boosts neurotransmitter norepinephrine/noradrenaline and may also increase dopamine in prefrontal cortex
- Blocks norepinephrine reuptake pump (norepinephrine transporter)
- Presumably, this increases noradrenergic neurotransmission
- Since dopamine is inactivated by norepinephrine reuptake in frontal cortex which largely lacks dopamine transporters, reboxetine can increase dopamine neurotransmission in this part of the brain

How Long Until It Works

- Onset of therapeutic actions usually not immediate, but often delayed 2–4 weeks
- If it is not working within 6–8 weeks for depression, it may require a dosage increase or it may not work at all
- May continue to work for many years to prevent relapse of symptoms

If It Works

- The goal of treatment is complete remission of current symptoms as well as prevention of future relapses
- Treatment most often reduces or even eliminates symptoms, but not a cure since symptoms can recur after medicine stopped
- Continue treatment until all symptoms are gone (remission)

- Once symptoms are gone, continue treating for 1 year for the first episode of depression
- For second and subsequent episodes of depression, treatment may need to be indefinite

If It Doesn't Work

- Many patients have only a partial response where some symptoms are improved but others persist (especially insomnia, fatigue, and problems concentrating)
- Other patients may be nonresponders, sometimes called treatment-resistant or treatment-refractory
- Consider increasing dose, switching to another agent, or adding an appropriate augmenting agent
- Consider psychotherapy
- Consider evaluation for another diagnosis or for a comorbid condition (e.g., medical illness, substance abuse, etc.)
- Some patients may experience apparent lack of consistent efficacy due to activation of latent or underlying bipolar disorder, and require antidepressant discontinuation and a switch to a mood stabilizer

Best Augmenting Combos for Partial Response or Treatment Resistance

- Trazodone, especially for insomnia

- SSRIs, SNRIs, mirtazapine (use combinations of antidepressants with caution as this may activate bipolar disorder and suicidal ideation)
- Modafinil, especially for fatigue, sleepiness, and lack of concentration
- Mood stabilizers or atypical antipsychotics for bipolar depression, psychotic depression, or treatment-resistant depression
- Benzodiazepines for anxiety
- Hypnotics for insomnia
- Classically, lithium, buspirone, or thyroid hormone

Tests

- None for healthy individuals

Side Effects

How Drug Causes Side Effects

- Norepinephrine increases in parts of the brain and body and at receptors other than those that cause therapeutic actions (e.g., unwanted actions of norepinephrine on acetylcholine release causing constipation and dry mouth, etc.)
- Most side effects are immediate but often go away with time

Notable Side Effects

- Insomnia, dizziness, anxiety, agitation
- Dry mouth, constipation
- Urinary hesitancy, urinary retention
- Sexual dysfunction (impotence)
- Dose-dependent hypotension



Life-Threatening or Dangerous Side Effects

- Rare seizures
- Rare induction of mania
- Rare activation of suicidal ideation and behavior (suicidality) (short-term studies did not show an increase in the risk of suicidality with antidepressants compared to placebo beyond age 24)

Weight Gain



- Reported but not expected

Sedation



- Reported but not expected

What to Do About Side Effects

- Wait

- Wait
- Wait
- Lower the dose
- In a few weeks, switch or add other drugs

Best Augmenting Agents for Side Effects

- For urinary hesitancy, give an alpha 1 blocker such as tamsulosin
- Often best to try another antidepressant monotherapy prior to resorting to augmentation strategies to treat side effects
- Trazodone or a hypnotic for drug-induced insomnia
- Benzodiazepines for drug-induced anxiety and activation
- Mirtazapine for drug-induced insomnia or anxiety
- Many side effects are dose-dependent (i.e., they increase as dose increases, or they reemerge until tolerance redevelops)
- Many side effects are time-dependent (i.e., they start immediately upon dosing and upon each dose increase, but go away with time)
- Activation and agitation may represent the induction of a bipolar state, especially a mixed dysphoric bipolar II condition sometimes associated with suicidal ideation, and require the addition of lithium, a mood stabilizer or an atypical antipsychotic, and/or discontinuation of reboxetine

Dosing and Use

Usual Dosage Range

- 8 mg/day in 2 doses (10 mg usual maximum daily dose)

Dosage Forms

- Tablet 2 mg, 4 mg scored

How to Dose

- Initial 2 mg/day twice a day for 1 week, 4 mg/day twice a day for second week



Dosing Tips

- When switching from another antidepressant or adding to another antidepressant, dosing may need to be lower and titration slower to prevent activating side effects (e.g., 2 mg in the daytime for 2–3 days, then 2 mg bid for 1–2 weeks)
- Give second daily dose in late afternoon rather than at bedtime to avoid undesired activation or insomnia in the evening
- May not need full dose of 8 mg/day when given in conjunction with another antidepressant
- Some patients may need 10 mg/day or more if well tolerated without orthostatic hypotension and if additional efficacy is seen at high doses in difficult cases

- Early dosing in patients with panic and anxiety may need to be lower and titration slower, perhaps with the use of concomitant short-term benzodiazepines to increase tolerability

Overdose

- Postural hypotension, anxiety, hypertension

Long-Term Use

- Safe

Habit Forming

- No

How to Stop

- Taper not necessary

Pharmacokinetics

- Metabolized by CYP450 3A4
- Inhibits CYP450 2D6 and 3A4 at high doses
- Elimination half-life approximately 13 hours



Drug Interactions

- Tramadol increases the risk of seizures in patients taking an antidepressant

- May need to reduce reboxetine dose or avoid concomitant use with inhibitors of CYP450 3A4, such as azole and antifungals, macrolide antibiotics, fluvoxamine, nefazodone, fluoxetine, sertraline, etc.
- Via CYP450 2D6 inhibition, reboxetine could theoretically interfere with the analgesic actions of codeine, and increase the plasma levels of some beta blockers and of atomoxetine and TCAs
- Via CYP450 2D6 inhibition, reboxetine could theoretically increase concentrations of thioridazine and cause dangerous cardiac arrhythmias
- Via CYP450 3A4 inhibition, reboxetine may increase the levels of alprazolam, buspirone, and triazolam
- Via CYP450 3A4 inhibition, reboxetine could theoretically increase concentrations of certain cholesterol lowering HMG CoA reductase inhibitors, especially simvastatin, atorvastatin, and lovastatin, but not pravastatin or fluvastatin, which would increase the risk of rhabdomyolysis; thus, coadministration of reboxetine with certain HMG CoA reductase inhibitors should proceed with caution
- Via CYP450 3A4 inhibition, reboxetine could theoretically increase the concentrations of pimozide, and cause QTc prolongation and dangerous cardiac arrhythmias
- Use with ergotamine may increase blood pressure
- Hypokalemia may occur if reboxetine is used with diuretics
- Use with caution with MAO inhibitors, including 14 days after MAOIs are stopped

Other Warnings/Precautions

- Use with caution in patients with bipolar disorder unless treated with concomitant mood-stabilizing agent
- Use with caution in patients with urinary retention, benign prostatic hyperplasia, glaucoma, epilepsy
- Use with caution with drugs that lower blood pressure
- When treating children, carefully weigh the risks and benefits of pharmacological treatment against the risks and benefits of nontreatment with antidepressants and make sure to document this in the patient's chart
- Distribute the brochures provided by the FDA and the drug companies
- Whenever possible, warn patients and their caregivers about the possibility of activating side effects, and advise them to report such symptoms immediately
- Monitor patients for activation of suicidal ideation, especially children and adolescents

Do Not Use

- If patient has angle-closure glaucoma
- If patient is taking an MAOI (except as noted under drug interactions)
- If patient is taking pimozide or thioridazine

- If there is a proven allergy to reboxetine

Special Populations

Renal Impairment

- Plasma concentrations are increased
- May need to lower dose

Hepatic Impairment

- Plasma concentrations are increased
- May need to lower dose

Cardiac Impairment

- Use with caution

Elderly

- Lower dose is recommended (4–6 mg/day)
- Reduction in the risk of suicidality with antidepressants compared to placebo in adults age 65 and older



Children and Adolescents

- Carefully weigh the risks and benefits of pharmacological treatment against the risks and benefits of nontreatment with antidepressants and make sure to document this in the patient's chart

- Monitor patients face-to-face regularly, particularly during the first several weeks of treatment
- Use with caution, observing for activation of known or unknown bipolar disorder and/or suicidal ideation, and inform parents or guardians of this risk so they can help observe child or adolescent patients
- No guidelines for children; safety and efficacy have not been established

Pregnancy

- No controlled studies in humans
- Not generally recommended for use during pregnancy, especially during first trimester
- Must weigh the risk of treatment (first trimester fetal development, third trimester newborn delivery) to the child against the risk of no treatment (recurrence of depression, maternal health, infant bonding) to the mother and child
- For many patients this may mean continuing treatment during pregnancy

Breast Feeding

- Some drug is found in mother's breast milk
- Immediate postpartum period is a high-risk time for depression, especially in women who have had prior depressive episodes, so

drug may need to be reinstituted late in the third trimester or shortly after childbirth to prevent a recurrence during the postpartum period

- Must weigh benefits of breast feeding with risks and benefits of antidepressant treatment versus nontreatment to both the infant and the mother
- For many patients, this may mean continuing treatment during breast feeding

The Art of Psychopharmacology

Potential Advantages

- Tired, unmotivated patients
- Patients with cognitive disturbances
- Patients with psychomotor retardation

Potential Disadvantages

- Patients unable to comply with twice daily dosing
- Patients unable to tolerate activation

Primary Target Symptoms

- Depressed mood
- Energy, motivation, and interest
- Suicidal ideation

- Cognitive disturbance
- Psychomotor retardation

 Pearls

- May be effective if SSRIs have failed or for SSRI “poop-out”
- * May be more likely than SSRIs to improve social and work functioning
- Reboxetine is a mixture of an active and an inactive enantiomer, and the active enantiomer may be developed in future clinical testing
- * Side effects may appear “anticholinergic,” but reboxetine does not directly block muscarinic receptors
- Constipation, dry mouth, and urinary retention are noradrenergic, due in part to peripheral alpha 1 receptor stimulation causing decreased acetylcholine release
- * Thus, antidotes for these side effects can be alpha 1 antagonists such as tamsulosin, especially for urinary retention in men over 50 with borderline urine flow
- Novel use of reboxetine may be for attention deficit disorder, analogous to the actions of another norepinephrine selective reuptake inhibitor, atomoxetine, but few controlled studies
- Another novel use may be for neuropathic pain, alone or in combination with other antidepressants, but few controlled studies
- Some studies suggest efficacy in panic disorder

Suggested Reading

Eyding D , Lelgemann M , Grouven U , et al. Reboxetine for acute treatment of major depression: systematic review and meta-analysis of published and unpublished placebo and selective serotonin reuptake inhibitor controlled trials . *BMJ* 2010 ;**341** :c4737 .

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Fleishaker JC . Clinical pharmacokinetics of reboxetine, a selective norepinephrine reuptake inhibitor for the treatment of patients with depression . *Clin Pharmacokinet* 2000 ;**39** (6):413–27 .

=====

Kasper S , el Giamal N , Hilger E . Reboxetine: the first selective noradrenaline re-uptake inhibitor . *Expert Opin Pharmacother* 2000 ;**1** (4):771–82 .

=====

Keller M . Role of serotonin and noradrenaline in social dysfunction: a review of data on reboxetine and the Social Adaptation Self-evaluation Scale (SASS) . *Gen Hosp Psychiatry* 2001 ;**23** (1):15 –19 .

=====

Tanum L . Reboxetine: tolerability and safety profile in patients with major depression . *Acta Psychiatr Scand Suppl* 2000 ;**402** :37 –40 .